The Ontological Neutrality Theorem: Structural Constraints for Neutral Substrates Under Persistent Interpretive Disagreement

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Abstract

Modern data systems must support accountability across persistent legal, political, and analytic disagreement regarding the interpretation of events, responsibilities, and explanations. This requirement imposes strict constraints on the design of any ontology intended to function as a shared substrate. This paper establishes a necessary and sufficient condition for ontological neutrality: neutrality, understood as interpretive non-commitment and extension stability across incompatible frameworks, holds if and only if causal or normative commitments are excluded from the substrate layer. It follows that neutral ontological substrates must be pre-causal and pre-normative at the foundational layer, representing entities together with their identity and persistence conditions, while externalizing causal and normative interpretation, evaluation, and explanation to extension layers. The paper does not propose a specific ontology or protocol; rather, it identifies the structural constraints that any ontology must satisfy if it is intended to function as a neutral substrate across conflicting interpretive frameworks.

Preview of Main Result. *Let $\mathcal{S}$ be a substrate ontology intended to function as a neutral substrate under diverse interpretive frameworks. Then $\mathcal{S}$ satisfies the requirements of neutrality if and only if its foundational Level of Abstraction does not assert causal or normative commitments as substrate-level facts.*

Informally. The result applies to systems that adopt the design objective of maintaining a neutral substrate under persistent interpretive disagreement. Under this objective, the theorem establishes that a neutral substrate must be pre-causal and pre-normative at the foundational layer. Causal and normative concepts may appear in the substrate's vocabulary, and causal and normative claims may be represented; what is excluded is asserting any such claim as a substrate-level commitment whose truth is fixed at the substrate layer independently of any interpretive framework. The precise conditions governing such extensions are stated in Section 3. Systems that enforce interpretive convergence through coordinated governance or shared authority, or that do not adopt the neutral-substrate design objective, are not subject to the constraints established here.

The formal statement and proof appear in Section 4.

Keywords: Formal ontology; ontological neutrality; accountability; neutrality constraints; extension stability; interpretive non-commitment; pre-causal; pre-normative

1 Introduction: Accountability, Disagreement, and Neutral Substrates

Modern data systems must support accountability across persistent legal, political, and analytic disagreement regarding the interpretation of events, responsibilities, and explanations. Legal interpretations diverge across jurisdictions and over time (Dworkin 1986; Hart 1961; Berman 2007); political actors contest responsibility and intent (Bovens 2007; Hood 2011); analytic frameworks disagree about causation and attribution (Woodward 2003). These disagreements are not anomalies to be resolved prior to data representation; they are structural features of institutional environments that accountability-oriented systems must accommodate while maintaining stable reference to shared entities.

Accountability, in this context, requires more than the collection of records or the publication of datasets (Fox 2007; Bovens et al. 2014). It requires the ability to refer stably to entities, occurrences, and institutional artifacts (Gruber 1993; Smith and Ceusters 2010), while allowing competing explanations, evaluations, and judgments to coexist. A system that collapses disagreement by embedding a single interpretation into its foundational representations restricts accountability in settings characterized by persistent interpretive disagreement, as it forecloses contestation and revision (Bowker and Star 1999; Winner 1980).

This paper adopts a different framing of the design challenge: rather than enforcing interpretive agreement or mediating semantic heterogeneity, the goal is to maintain stable shared reference across incompatible interpretations.

Design Problems in Shared Representation

Ontology design has traditionally addressed two related but distinct representation problems. First, systems may assume *shared interpretation*. In this setting, participants agree on a common ontology that provides a unified interpretation of shared representations. Second, systems may face *semantic heterogeneity*. Here multiple ontologies exist, but disagreement is treated as a translation problem addressed through alignment, mapping, or mediation mechanisms.

The problem considered in this paper is different. Many accountability-oriented environments exhibit *persistent interpretive disagreement*. Legal, political, and analytic frameworks may reach incompatible conclusions about the same entities and events while continuing to operate over a shared representational base.

Ontology Design Strategies Across Different Semantic Conditions

Setting	**Ontology Structure**	**Mechanism**
Shared interpretation	shared ontology	unified interpretation
Semantic heterogeneity	multiple ontologies	alignment / mediation
Persistent disagreement	neutral substrate	multiple frameworks

Traditional ontology frameworks regulate how agreement is structured across systems. Neutral substrate architectures instead regulate how disagreement can coexist without forcing revision of the shared representational base.

This paper examines the implications of this requirement for ontology design. Specifically, it addresses the conditions under which an ontology can function as a *neutral substrate* for accountability: a shared representational layer that supports stable reference to entities, occurrences, and institutional artifacts while allowing multiple incompatible interpretive frameworks to coexist. Under this conception of neutrality, the substrate must remain usable across divergent legal, political, and analytic frameworks without requiring revision of the substrate itself. A neutral substrate, in this sense, is a foundational representation that supports such coexistence without embedding the causal or normative conclusions on which those frameworks may disagree.

In this paper, a *substrate ontology* is an ontology that serves as the shared representational base across interpretive frameworks. Its foundational layer provides the entities and relations over which interpretive frameworks reason and derive conclusions. For brevity, this ontology may be referred to as the *substrate*.

This requires a careful selection of the Level of Abstraction (LoA) (Floridi 2008) at which the substrate operates. Following Floridi, an LoA is defined by the set of observables it makes available for representation. In an ontological substrate, these observables correspond to the predicate symbols available in the foundational vocabulary. For a substrate to remain neutral, it must operate at an LoA whose vocabulary excludes explanatory or evaluative predicates.

Conceptually, the architecture considered here distinguishes between several representational roles, defined precisely in Section 3. *Logical propositions* are the representational contents under consideration: statements about entities, events, causal relations, and normative attributions. An *ontology* asserts a subset of those propositions as *commitments*: claims that hold at the substrate level independently of any particular interpretive framework. An *interpretive framework* is a set of rules or principles that reasons over substrate commitments and derives *conclusions*: causal attributions, responsibility assignments, or normative evaluations that are not asserted by the substrate itself.

A *neutral substrate* constrains which propositions may be asserted as commitments so that multiple frameworks may derive incompatible conclusions from the same substrate without requiring revision of the substrate. The key design question is therefore which propositions may be safely asserted as substrate-level commitments without committing the substrate to any particular interpretive framework.

This form of neutrality imposes strict design constraints. In particular, ontologies that assert causal or normative commitments at the foundational layer cannot satisfy the requirements of interpretive non-commitment and extension stability that neutrality demands. This constraint is consistent with prior observations that classification systems and representational structures embed normative and interpretive commitments (Bowker and Star 1999; Hacking 1999). Causal and normative conclusions may be derived by interpretive frameworks, but embedding such conclusions as substrate-layer commitments commits the substrate to a particular framework and therefore violates neutrality.

This paper adopts the following design constraint. Under persistent disagreement, extensions to the substrate may introduce incompatible causal accounts or normative attributions, and those extensions are not required to converge. Substrate revision is treated as an impermissible mechanism for resolving interpretive conflict: the substrate must remain stable across extensions whose conclusions are mutually inconsistent. This is a deliberate design choice, not a universal requirement. Systems that can enforce interpretive convergence through coordinated governance or shared authority do not face this constraint; the results of this paper apply only to substrates explicitly designed to remain stable without such coordination.

The contribution of this paper is not the observation that interpretive frameworks may disagree about causal or normative conclusions. That fact is widely recognized across legal theory, political analysis, and causal modeling. The contribution is to show that, in settings where substrate revision is treated as an impermissible mechanism for resolving interpretive conflict, this disagreement imposes structural constraints on ontology design, and to establish those constraints formally. Neutrality is formalized as two requirements: *interpretive non-commitment* and *extension stability*. The central result is a biconditional. In the necessity direction, any substrate that embeds causal or normative commitments at the foundational layer will be inconsistent with some admissible interpretive framework, violating both requirements. In the sufficiency direction, a substrate that restricts its foundational layer to framework-invariant referential structure with explicit identity and persistence conditions can satisfy both requirements simultaneously, without introducing hidden commitments through role predicates or contextual typing. Neutrality is therefore possible if and only if the substrate is pre-causal and pre-normative. Causal and normative propositions may not be asserted as substrate commitments but may be introduced within interpretive frameworks layered above the substrate. The precise conditions under which extensions may be introduced are stated in Section 3.

The remainder of the paper proceeds as follows. Section 2 reviews related work in ontology design, legal informatics, and causal modeling, situating the present contribution. Section 3 defines the notion of a neutral substrate and specifies the two neutrality requirements relevant to accountability systems. Section 4 establishes the central result: a neutral substrate satisfies both requirements if and only if its foundational layer excludes causal and normative commitments. Section 5 discusses the implications of this result for ontology design, clarifying what must be excluded from foundational layers and what may be externalized to interpretive frameworks. Section 6 summarizes the resulting constraints and discusses their relevance for ontology design in accountability contexts.

2 Related Work

The constraints developed in this paper intersect with several lines of prior work: upper ontologies in formal ontology research, domain ontologies for legal and institutional reasoning, and causal modeling frameworks. This section situates the present contribution relative to each, clarifying both debts and departures.

2.1 Political Philosophy: Reasonable Pluralism

The premise that disagreement is structural rather than resolvable draws on Rawls' concept of reasonable pluralism (Rawls 1993). Rawls argues that under conditions of free inquiry, reasonable persons will arrive at incompatible comprehensive doctrines.

This paper transposes that insight to ontology design: if interpretive disagreement is a permanent feature of accountability contexts, then neutral substrates must accommodate it structurally rather than resolving it by fiat. The central result formalizes this accommodation as a structural constraint on substrate-level commitments. Rawls' argument is not imported as a normative foundation, but as an analogy for the persistence of disagreement that motivates the need for structural accommodation.

2.2 Upper Ontologies: BFO and DOLCE

The Basic Formal Ontology (BFO) (International Organization for Standardization and International Electrotechnical Commission 2021; Arp et al. 2015) and the Descriptive Ontology for Linguistic and Cognitive Engineering (DOLCE) (Borgo et al. 2022a; Masolo et al. 2003) represent two influential approaches to foundational ontology design. Both aim to provide domain-independent upper-level categories that can be extended for specific applications. A recent special issue (Borgo et al. 2022b) provides systematic comparison of seven foundational ontologies (BFO, DOLCE, GFO, GUM, TUpper, UFO, YAMATO) through common modeling cases, illustrating both their shared commitments and divergent design choices.

BFO distinguishes continuants from occurrents and provides a realist framework grounded in scientific practice. DOLCE takes a more cognitive and linguistic orientation, emphasizing the role of conceptualization in ontological commitment. Both have been widely adopted in biomedical, engineering, and information systems contexts.

The present work addresses a different question: it analyzes the logical consequences of embedding interpretive commitments at the substrate level, rather than the adequacy or intent of any particular foundational ontology. BFO and DOLCE provide categorical structure for what exists; they do not explicitly address the conditions under which an ontology can remain neutral across interpretive disagreement. These frameworks may be extended in practice to include causal and normative relations as first-class ontological commitments. The result established here implies that such extensions, if embedded at the substrate layer, compromise neutrality in the sense required for accountability systems.

In Floridi's terms, while BFO and DOLCE provide foundational ontological structure, they do not mandate a specific Level of Abstraction that excludes interpretive predicates. Our contribution is to show that for the specific purpose of neutrality, such a mandate is a logical necessity.

Guarino's analysis of ontological commitment (Guarino et al. 2009) clarifies what it means for an ontology to assert something; the present contribution identifies which such assertions are incompatible with neutrality.

This paper's constraints are therefore compatible with BFO or DOLCE as upper-level scaffolding, provided that causal and normative commitments are externalized to interpretive layers rather than embedded in the foundational substrate. Recent work on formal alignment between BFO and DOLCE (Masolo et al. 2025) demonstrates that even foundational ontologies with divergent commitments can be mapped systematically, reinforcing the feasibility of neutral substrates that support multiple interpretive frameworks.

Whether specific versions or extensions of BFO, DOLCE, or other upper ontologies satisfy these constraints in practice is an empirical question beyond the scope of this paper. Such compatibility assessments require detailed analysis of particular ontology versions and their commitments, work that would benefit from collaboration with the communities that maintain and extend these frameworks.

2.3 Legal Ontologies: LKIF and Related Approaches

Legal ontologies, e.g., the Legal Knowledge Interchange Format (LKIF) (Hoekstra et al. 2007), and related frameworks (Sartor et al. 2011; Breuker and Hoekstra 2009) aim to represent legal concepts, norms, and reasoning patterns in machine-processable form. These ontologies typically include deontic categories, such as obligation, permission, and prohibition, as core primitives, along with relations for legal causation, responsibility, and compliance.

Such frameworks are valuable for legal reasoning and case analysis within a fixed normative framework. However, they are not designed to function as neutral substrates across incompatible legal or political interpretations. By embedding deontic conclusions as ontological facts at the foundational layer, such designs commit the ontology to a particular normative stance. When legal interpretations diverge, as they routinely do across jurisdictions, over time, or under political contestation, the substrate ontology itself must be revised to accommodate disagreement.

The result established here applies directly to such designs. LKIF and similar ontologies are appropriate for interpretive layers built atop a neutral substrate, but they cannot themselves serve as the substrate if neutrality across disagreement is required.

2.4 Causal Modeling: Pearl and Structural Approaches

The structural causal modeling framework developed by Pearl (Pearl 2009) and extended by others (Woodward 2003; Spirtes et al. 2000) provides rigorous tools for representing and reasoning about causation. The do-calculus and related interventionist semantics have become standard in causal inference across multiple disciplines.

These frameworks make explicit causal commitments: they assert that certain variables cause others under specified structural assumptions. Such commitments are essential for causal reasoning but are inherently model-dependent. Different causal models, each internally consistent, may disagree about which relations are causal, which variables are confounders, and which interventions produce which effects.

The present work does not dispute the value of causal modeling for explanation and decision-making. Rather, it observes that causal conclusions are framework-dependent in the same structural sense as normative conclusions and formalized in Lemma 4.1. Embedding causal relations as substrate-layer primitives commits the ontology to a particular causal model, violating interpretive non-commitment. Causal reasoning must therefore be externalized to interpretive layers where model assumptions are explicit and disagreement can be represented without substrate revision.

2.5 Philosophy of Information: Levels of Abstraction

The methodological framework of this paper is informed by Floridi's Philosophy of Information (Floridi 2008 2011). Floridi introduces the Method of Levels of Abstraction (LoA) as a way to specify the epistemic range of a system through a set of observables. In ontological terms, these observables correspond to the predicate symbols available in the system's representational vocabulary. Crucially, Floridi distinguishes between the LoA (the lens through which a system is viewed) and the Level of Organization (LoO), which refers to the internal structural hierarchy of the system itself.

While Floridi uses these concepts to analyze the nature of data and knowledge, this paper applies them specifically to the problem of ontological neutrality. Under this view, neutrality is a function of operating at a foundational LoO with a deliberately restricted LoA. This paper extends Floridi's work by proving that causal and normative predicates cannot be asserted at the substrate layer without violating neutrality under extension by admissible interpretive frameworks.

2.6 Information Infrastructure: Boundary Objects

Star and Griesemer's concept of boundary objects (Star 1989; Star and Ruhleder 1996) and subsequent work on classification systems (Bowker and Star 1999) address how representations can be shared across communities with divergent interpretive frameworks. Boundary objects are plastic enough to adapt to local needs yet robust enough to maintain identity across sites.

The neutral substrate concept developed here can be understood as a formalization of this insight: the substrate provides the shared identity structure, while interpretive frameworks supply local meaning. The theorem specifies what cannot be shared at the boundary level without losing plasticity.

2.7 Positioning the Present Contribution

The contribution of this paper is a meta-theoretical constraint. Unlike BFO and DOLCE, it does not propose a specific taxonomy of entities; rather, it defines the informational boundary that any such taxonomy must respect to remain neutral.

Recent work has questioned the stability of top-level ontological commitments in emerging technical domains. Köhler and Neuhaus argue that large language models exhibit a *mercurial* ontological status: their classification shifts depending on task framing, deployment context, and explanatory purpose (Köhler and Neuhaus 2025). This observation is consistent with the present result by illustrating that even top-level ontological categories cannot be guaranteed to remain stable when interpretive predicates are embedded, reinforcing the need, where neutrality across uses is required, for a neutrality-preserving substrate beneath such shifting conceptualizations.

Recent work also emphasizes that ontology development itself requires negotiating consensus among stakeholders with divergent commitments (Neuhaus and Hastings 2022). This observation reinforces the present argument by contrast: while temporary disagreement during ontology construction may be instrumental in converging on a sound design, ontologies intended to operate after construction in contexts of valid and persistent interpretive disagreement among admissible frameworks cannot rely on consensus as a foundational design assumption. Instead, such ontologies must, in these contexts, be structured to accommodate disagreement structurally without requiring its resolution. The theorem specifies the formal constraints such accommodation must satisfy.

Recent ontology engineering work has emphasized tool-supported maintenance and rapid evolution in fast-moving domains. For example, the Artificial Intelligence Ontology (AIO) employs LLM-assisted curation and automated ODK pipelines to keep a domain ontology current as AI concepts and ethical concerns evolve (Joachimiak et al. 2025). Such approaches highlight the importance of distinguishing between evolving domain-level ontologies and the comparatively stable substrates on which they depend: the present result clarifies why neutrality constraints must apply to the latter even as the former necessarily change.

By grounding this constraint in the Method of Levels of Abstraction, the debate shifts from a sociological discussion of agreement to a formal requirement for observational invariance.

This result is complementary to prior work rather than competitive. Upper ontologies may inform the categorical structure of a neutral substrate. Legal and causal frameworks may operate as interpretive layers atop it. What the present work provides is a principled boundary: a specification of what must be externalized in order for the substrate to support accountability across disagreement.

3 Formal Requirements for Ontological Neutrality

This section formalizes the two neutrality requirements introduced in Section 1. All formal results in subsequent sections are conditional on the definitions stated here. Unless otherwise noted, all entailment ($\vdash$) is with respect to classical first-order logic with equality.

The conceptual architecture is summarized in Figure 1.

The interface between layers consists of shared reference to entities, occurrences, and institutional artifacts, together with their identity and persistence conditions; interpretive frameworks extend this structure without revising substrate-layer commitments.

The formal requirements governing this architecture are stated in the subsections that follow.

The notion of neutrality used in this paper is deliberately narrow and operational. It does not refer to moral neutrality, political impartiality, or

Interpretive Frameworks
(causal and normative reasoning; competing conclusions)

$\uparrow$ extends, but does not revise substrate commitments

Neutral Substrate
Framework-invariant referential structure:
identity and persistence conditions for
entities, occurrences, and institutional artifacts

Figure 1: Conceptual architecture of a neutral substrate and interpretive frameworks.

epistemic skepticism. Rather, neutrality is defined relative to the functional role an ontology plays when it is used as a shared substrate across divergent interpretive frameworks.

An ontological substrate is neutral, in the sense relevant here, if it satisfies two requirements. These requirements are independent but jointly necessary. Together, they characterize the minimal conditions under which a shared ontology can support accountability without foreclosing disagreement.

Throughout this section, all requirements are understood as analytic consequences of this role-relative definition of neutrality, not as general design prescriptions for ontologies.

Interpretive non-commitment constrains what the substrate asserts at its chosen Level of Abstraction. Extension stability is a global consistency condition: even a substrate that is silent on contested predicates can fail neutrality if it encodes background axioms that conflict with some admissible interpretive framework.

3.1 Interpretive Non-Commitment

The first requirement, under this definition of neutrality, is interpretive non-commitment. A substrate ontology satisfies this requirement if it does not assert any proposition whose truth value can vary across admissible interpretive frameworks.

Admissibility is determined by the domain of application, not by the substrate. A framework is admissible if recognized stakeholders in the domain accept it as a legitimate interpretive stance, for example, a jurisdiction's legal code, a scientific community's causal methodology, or an institution's normative charter. The substrate must accommodate all such frameworks; it does not adjudicate among them.

Definition 3.1 (Admissible Framework)**.** A framework $\mathcal{F}$ is *admissible* if it is internally consistent ($\mathcal{F} \nvdash \bot$) and represents a legitimate interpretive stance within the domain of application. The set of all admissible frameworks is denoted $\mathbb{F}$. Admissible frameworks need not be mutually compatible.

With this notion, the first requirement can be stated precisely.

Definition 3.2 (Framework-Variant Proposition)**.** Let $\mathcal{S}$ denote the substrate ontology whose neutrality properties are under analysis. A proposition p whose truth depends on the interpretive framework is *framework-variant* if there exist admissible frameworks $\mathcal{F}_1, \mathcal{F}_2 \in \mathbb{F}$ such that $\mathcal{S} \cup \mathcal{F}_1 \vdash p$ and $\mathcal{S} \cup \mathcal{F}_2 \vdash \neg p$.

Definition 3.3 (Interpretive Non-Commitment)**.** A substrate ontology $\mathcal{S}$ satisfies *interpretive non-commitment* if it does not assert any framework-variant proposition: there is no proposition p such that $\mathcal{S} \vdash p$ and p is framework-variant with respect to $\mathbb{F}$.

Accountability systems routinely involve disagreements about what occurred, why it occurred, and how it should be evaluated. Legal frameworks may disagree about whether an action was permitted or prohibited; analytic frameworks may disagree about whether one event caused another; political frameworks may disagree about responsibility or intent. These disagreements are not merely epistemic gaps to be filled with additional data. They reflect differences in background assumptions, governing rules, and interpretive commitments.

A neutral substrate, under this definition, must therefore refrain from settling such questions at the ontological level. If a substrate ontology asserts that a particular action was obligatory, forbidden, or permitted, it commits the ontology to a particular normative framework. Similarly, if it asserts that one event caused another, it commits the ontology to a particular causal model or explanatory framework. This does not imply that causation is subjective or unreal. Rather, causal conclusions depend on explanatory frameworks (e.g., statistical, mechanistic, or legal standards of causation) that may legitimately differ across admissible frameworks. In both cases, asserting such conclusions as substrate-layer commitments violates interpretive non-commitment and therefore neutrality.

Interpretive non-commitment does not require the substrate ontology to ignore normative or causal discourse altogether. Rather, it requires that the substrate operate at a Level of Abstraction (LoA) (Floridi 2008 2011) that excludes explanatory or evaluative predicates, including causal and normative predicates.

In this framework, an LoA specifies the set of predicate symbols that define the referential structure available at that level. Neutrality is achieved by selecting an LoA in which these predicates are restricted to entity identity, and persistence conditions (i.e., how entities are individuated and tracked across time).

This referential structure constitutes the invariant core of the representation, remaining stable under extension by admissible interpretive frameworks regardless of the higher-level interpretive logic applied to them. Claims about obligations, permissions, or causation are thus represented as external assertions made by agents within an interpretive framework, rather than being internalized as substrate-layer facts.

3.2 Extension Stability

The second requirement, given the neutrality role defined above, is extension stability. A substrate ontology satisfies this requirement if it remains consistent when extended by incompatible interpretive frameworks.

Definition 3.4 (Extension Stability)**.** A substrate ontology $\mathcal{S}$ satisfies *extension stability* if for all admissible frameworks $\mathcal{F} \in \mathbb{F}$, $\mathcal{S} \cup \mathcal{F} \nvdash \bot$.

This requirement follows from the functional role of a neutral substrate, which must remain usable by independently developed interpretive frameworks whose conclusions may diverge and whose development cannot be centrally coordinated.

Extension stability does not require that all admissible interpretive frameworks be simultaneously combined into a single, globally consistent theory. Rather, the requirement is *pairwise compatibility with the substrate*: for each admissible framework $\mathcal{F} \in \mathbb{F}$ considered independently, the combined theory $\mathcal{S} \cup \mathcal{F}$ must remain consistent. The substrate is permitted to be extended by additional structures, records, or interpretive layers, provided such extensions do not retract, revise, or contradict substrate-layer assertions. A violation occurs only when accommodating a new admissible framework requires modification of the substrate itself. Disagreement between frameworks, including disagreement about causal, normative, or explanatory conclusions, is expected and permitted; what is prohibited is embedding substrate-layer commitments that require revision when such disagreement arises.

In practice, interpretive frameworks evolve over time: legal interpretations change, analytic models are revised, and political judgments are contested. A neutral substrate must accommodate such developments without requiring revision of its foundational commitments. Disagreement should therefore produce additional interpretive layers rather than modification of the substrate itself.

Extension stability is violated when the introduction of a new interpretive framework requires retracting or revising ontological assertions.

Consider a substrate ontology $\mathcal{S}$ used by two interpretive frameworks, $\mathcal{F}_1$ and $\mathcal{F}_2$, that contradict each other ($\mathcal{F}_1 \cup \mathcal{F}_2 \vdash \bot$). For $\mathcal{S}$ to remain stable, it must be separately consistent with each framework: $\mathcal{S} \cup \mathcal{F}_1 \nvdash \bot$ and $\mathcal{S} \cup \mathcal{F}_2 \nvdash \bot$. The substrate need not reconcile the frameworks with each other; it must simply avoid assertions that either framework rejects.

Now suppose $\mathcal{S}$ embeds a causal or normative commitment c that $\mathcal{F}_1$ accepts but $\mathcal{F}_2$ rejects (i.e., $\mathcal{F}_2 \vdash \neg c$). Then $\mathcal{S} \cup \mathcal{F}_2 \vdash c \wedge \neg c \vdash \bot$: the combination is inconsistent, and $\mathcal{S}$ must be revised.

This demonstrates that embedding any causal or normative commitment at the substrate layer, regardless of current consensus, introduces latent instability. Such a substrate cannot function as a permanent neutral foundation, because some admissible framework may eventually reject what the substrate asserts.

This requirement rules out approaches in which neutrality is achieved through periodic revision or replacement of ontological commitments. Stability under extension is a necessary condition for interoperability across time, jurisdictions, and analytic communities.

3.3 Scope of Neutrality

It is important to emphasize that neutrality, as defined here, is not global. A neutral substrate is not neutral with respect to identity and persistence conditions. It must take substantive positions on how entities are individuated, what criteria govern their persistence, and therefore what counts as an entity within the substrate. These commitments are necessary for accountability and do not vary across admissible interpretive frameworks in the same way that causal or normative conclusions do.

Neutrality is therefore scoped specifically to interpretive commitments: those aspects of representation whose truth depends on legal, normative, causal, or analytic frameworks that may legitimately disagree. This restriction concerns what the substrate may assert, not whether causal or normative relations exist in the world. Such relations may be represented and reasoned about within interpretive frameworks layered above the substrate. The exclusion of such commitments from the substrate is not a matter of preference or conservatism; it is a functional requirement derived from the role the ontology is intended to play as a neutral substrate across divergent interpretive frameworks.

The design of a neutral substrate requires distinguishing between the LoA and the Level of Organization (LoO). Formally, the LoA of the substrate is identified with the set of predicate symbols available at the foundational layer. Neutrality requires that this set be restricted to *framework-invariant referential structure*: predicate symbols governing entity identity and persistence conditions. Predicate symbols whose extension depends on causal or normative interpretation are excluded from the foundational LoA but remain available to interpretive frameworks operating at higher organizational layers. The substrate records entities and occurrences, while causal and normative interpretation is deferred to extensions.

The LoO characterizes the architectural hierarchy of the system: the ordering of layers by direction of dependence, such that higher layers may extend but not revise lower ones. The neutral substrate occupies the primary LoO, providing the entity and identity conditions upon which higher-level interpretive layers are organized. Identity is established at the substrate layer before any interpretive framework is applied; explanation and interpretation operate on entities whose identity and persistence conditions are already fixed. This ensures that changes in higher-level organizational logic do not propagate downward to revise the foundational substrate.

A further scope condition applies. The neutrality requirements assume that identity and persistence conditions are themselves invariant across admissible frameworks. If frameworks disagree about identity, for example, whether two references denote the same actor, or whether a particular institutional artifact should be treated as a distinct entity, then the substrate cannot be neutral with respect to those frameworks. In such cases, identity must be resolved at a prior stage, or the domain does not admit a neutral substrate in the sense defined here. This is not a defect of the framework but a boundary condition: neutrality presupposes a shared ontological ground on which interpretation can vary.

Together, interpretive non-commitment and extension stability define the functional boundaries of a neutral substrate. These requirements establish that neutrality is not a lack of content, but a categorical constraint: to remain stable under pluralistic interpretation, a substrate ontology satisfies both requirements if and only if it is strictly pre-causal and pre-normative.

The next section establishes the central result, identifying the structural constraints a substrate ontology must satisfy to function as a neutral substrate across divergent interpretive frameworks.

4 Formal Development of the Ontological Neutrality Theorem

Section 3 defined the two neutrality requirements governing a substrate ontology: interpretive non-commitment and extension stability. This section establishes the relationship between those requirements and the structure of the substrate layer.

Specifically, it proves that a substrate satisfies the neutrality requirements if and only if causal and normative commitments are excluded from the substrate layer. The argument is structural and logical rather than empirical: it follows from the interaction between interpretive disagreement and substrate-level commitments. All necessity and sufficiency claims in this section are relative to the definitions introduced in Section 3.

4.1 Neutrality as Compatibility Across Admissible Frameworks

By definition, a neutral substrate must remain compatible with multiple admissible interpretive frameworks, even when those frameworks disagree. Compatibility here is understood in the sense of extension stability: the substrate must remain logically consistent when extended by each admissible interpretive framework, without revision or retraction of substrate-layer assertions.

A framework is *admissible* if it is internally consistent and represents a legitimate interpretive stance within the domain of application. Admissible interpretive frameworks may disagree about causal or normative conclusions concerning the same entities or events, while remaining internally consistent. For example, an admissible framework may correspond to a recognized legal jurisdiction, an established causal modeling tradition, or a coherent normative theory. Admissibility is not universality: admissible frameworks may contradict one another, as when two jurisdictions reach opposite legal conclusions or two causal models attribute responsibility differently. The set of admissible frameworks $\mathbb{F}$ is thus characterized by internal consistency, not mutual compatibility.

This neutrality requirement immediately constrains the Level of Abstraction (LoA) at which the substrate, under the neutrality definition, must operate. For a substrate to be neutral, its foundational commitments must remain invariant across all admissible interpretive frameworks. If the substrate asserts a proposition whose truth depends on a particular causal or normative interpretation, then there may exist admissible frameworks that derive both that proposition and its negation. Because neutrality requires the substrate to remain logically consistent when extended by every admissible framework, asserting such a proposition at the substrate layer necessarily produces inconsistency with at least one admissible framework extension. The neutrality condition is therefore violated.

This restriction does not prohibit the representation of causal or normative concepts themselves. A neutral substrate may contain vocabulary for representing causal or normative claims, provided those claims are represented as interpretive assertions rather than asserted as substrate-layer facts. What neutrality excludes is asserting a causal or normative proposition as a substrate-level commitment, since such propositions may be rejected by some admissible framework.

4.2 Framework-Contestability and Substrate Incompatibility

The arguments in the following subsections share a common structure, stated here as a general lemma.

Lemma 4.1 (Framework-Contestability Lemma)**.** *Let $\mathcal{S}$ be a substrate ontology intended to satisfy the neutrality requirements, and let p be a proposition whose truth conditions depend on the conclusions of an interpretive framework rather than on framework-invariant referential structure. Then $\mathcal{S}$ cannot assert p as a substrate-layer commitment without violating either interpretive non-commitment or extension stability.*

Proof. Since p is framework-dependent, there exist admissible frameworks $\mathcal{F}_1, \mathcal{F}_2 \in \mathbb{F}$ such that $\mathcal{F}_1 \vdash p$ and $\mathcal{F}_2 \vdash \neg p$. If $\mathcal{S} \vdash p$, then p is framework-variant (Definition 3.2), violating interpretive non-commitment. Furthermore, $\mathcal{S} \cup \mathcal{F}_2 \vdash (p \wedge \neg p)$, hence $\vdash \perp$, violating extension stability. $\square$

The two subsections that follow instantiate this lemma for normative and causal commitments respectively, showing in each case that the relevant class of propositions is framework-dependent in the sense required.

4.3 Normative Commitments

Normative conclusions are framework-dependent in the sense of Lemma 4.1. Their truth conditions depend on legal regimes, institutional authorities, temporal scope, and interpretive stance. It is therefore a routine and expected feature of accountability systems that admissible frameworks disagree about normative conclusions.

By Lemma 4.1, asserting any such proposition as a substrate-layer commitment thereby commits the substrate to a particular normative framework. Any framework that denies the asserted commitment becomes incompatible with the substrate, as it cannot be layered atop it without contradiction or revision. This incompatibility is not a matter of missing context or insufficient detail; it arises from the ontological act of asserting a contested normative commitment as a foundational fact of the substrate. As a result, no substrate that asserts normative commitments at the foundational layer can satisfy interpretive non-commitment, and when alternative normative interpretations arise, as they inevitably do, extension stability is violated.

4.4 Causal Commitments

Causal attributions are framework-dependent in the sense of Lemma 4.1. Claims such as e_1 *caused* e_2 depend on background assumptions about causal mechanisms, variable selection, counterfactual reasoning, and model scope. Distinct causal frameworks may be equally admissible while disagreeing about specific causal relationships.

By Lemma 4.1, asserting any causal relation as a substrate-layer commitment necessarily commits the substrate to one causal model among many. Frameworks that reject that model cannot be layered onto the substrate without conflict. Importantly, this argument does not rely on the presence of an active dispute over a specific event. Causal attribution cannot be asserted as a substrate-layer commitment without committing to a particular counterfactual or mechanistic logic: as soon as a substrate asserts $\mathsf{Caused}(e_1, e_2)$, it is incompatible with any framework that treats e_1 and e_2 as merely correlated or as effects of a common cause z.

To be clear, *pre-causal does not mean acausal.* The substrate does not deny that causation exists or that causal reasoning is valuable. Rather, it refrains from embedding causal conclusions as foundational commitments, reserving causal attribution for interpretive layers where framework assumptions are explicit and disagreement is representable. Causation is externalized, not eliminated.

The following example illustrates these three cases.

Example 4.1 (Interpretive Disagreement in Incident Investigation)**.** Consider a substrate $\mathcal{S}$ representing a workplace incident. The substrate asserts only framework-invariant referential structure:

- $\mathsf{Agent}(a)$: a is an agent
- $\mathsf{Occurrence}(e)$: e is an occurrence
- $\mathsf{Participant}(a, e)$: a participated in e
- $\mathsf{Before}(e_1, e_2)$: occurrence e_1 preceded e_2

The substrate asserts: $\mathsf{Agent}(a)$, $\mathsf{Occurrence}(e_1)$, $\mathsf{Occurrence}(e_2)$, $\mathsf{Participant}(a, e_1)$, $\mathsf{Before}(e_1, e_2)$.

Two admissible interpretive frameworks extend $\mathcal{S}$:

$\mathcal{F}_1$ (engineering analysis): applies a mechanistic causal model and derives $\mathsf{Caused}(e_1, e_2)$ and $\mathsf{Responsible}(a, e_2)$.

$\mathcal{F}_2$ (legal framework): applies jurisdictional standards of contributory negligence and derives $\neg\mathsf{Responsible}(a, e_2)$.

Since $\mathcal{F}_1 \vdash \mathsf{Responsible}(a, e_2)$ and $\mathcal{F}_2 \vdash \neg\mathsf{Responsible}(a, e_2)$, the frameworks are mutually inconsistent: $\mathcal{F}_1 \cup \mathcal{F}_2 \vdash \bot$. However, each is separately consistent with $\mathcal{S}$: $\mathcal{S} \cup \mathcal{F}_1 \nvdash \bot$ and $\mathcal{S} \cup \mathcal{F}_2 \nvdash \bot$. The substrate remains neutral.

Violation case. Suppose instead $\mathcal{S}$ embeds the causal commitment $\mathsf{Caused}(e_1, e_2)$ at the foundational layer. Then $\mathcal{S} \cup \mathcal{F}_2$ must accommodate both $\mathsf{Caused}(e_1, e_2)$ (from $\mathcal{S}$) and the legal framework's background axiom (in this instance) that mere temporal precedence without established mechanism does not constitute causation. These commitments are inconsistent under that framework: $\mathcal{S} \cup \mathcal{F}_2 \vdash \bot$. To restore consistency, $\mathcal{S}$ must be revised, violating extension stability.

Reification case. Now suppose $\mathcal{S}$ neither asserts nor denies $\mathsf{Caused}(e_1, e_2)$, but represents the engineering framework's causal attribution as a reified assertion: $\mathsf{Asserts}(\mathcal{F}_1, \mathsf{Caused}(e_1, e_2))$. This is a substrate-level fact about the existence of an assertion, not a commitment to its content. Both $\mathcal{F}_1$ and $\mathcal{F}_2$ can be layered atop $\mathcal{S}$ without contradiction: each framework reasons over the same entities and occurrences while deriving its own conclusions. This is the pattern of DOLCE/DUL-style situation modeling: causal and normative claims are represented as situation-bound descriptions attributed to agents or frameworks, not as substrate-layer facts. Such modeling satisfies the structural constraint identified by the theorem precisely because it externalizes rather than embeds interpretive commitments.

4.5 Reification and Contextual Modeling

The reification case in Example 4.1 illustrates a broader point about contextual modeling approaches. Representing causal or normative claims as reified assertions attributed to agents or frameworks, rather than asserting them as substrate-layer facts, is consistent with the structural constraint identified here. Such approaches externalize interpretation rather than embedding it, and therefore satisfy the neutrality requirements.

This pattern is exemplified by DOLCE/DUL-style modeling of situations and descriptions (Borgo et al. 2022a). In this approach, causal or normative claims are represented as situation-bound descriptions: entities in the substrate denote the existence of an assertion or description, not the truth of its content. Different agents or frameworks may hold incompatible descriptions of the same occurrence without contradiction at the substrate level, because the substrate asserts only that such descriptions exist and are attributed to specific sources, not that their contents hold as ontological facts. This is precisely the externalization the theorem requires.

Reification preserves interpretive non-commitment only if the substrate refrains from simultaneously asserting the reified content as a substrate-layer truth. If causal or normative relations are asserted as substrate-layer facts and reified as claims, neutrality is already lost. If they are represented solely as reified claims attributed to interpretive frameworks, they function as objects of discourse layered atop the substrate rather than as foundational commitments.

Thus, reification and contextual modeling do not circumvent the theorem's constraint: they satisfy it, provided they are used to externalize interpretive commitments rather than to embed them under a different description. The constraint concerns what is asserted as foundational fact, not the representational mechanism through which interpretive discourse is recorded.

The preceding analysis establishes that causal and normative commitments are framework-variant and that neutrality requires substrate assertions to remain compatible with every admissible framework extension; the following theorem formalizes the resulting constraint on substrate design.

4.6 The Ontological Neutrality Theorem

The preceding observations yield the central result of this paper.

Theorem 4.2 (Ontological Neutrality Theorem)**.** *Let $\mathcal{S}$ be a substrate ontology intended to function as a neutral substrate across diverse interpretive frameworks, and let $\mathbb{F}$ be the set of admissible interpretive frameworks. Then $\mathcal{S}$ satisfies the requirements of neutrality (interpretive non-commitment and extension stability) if and only if its foundational Level of Abstraction does not assert causal or normative commitments as substrate-level ontological facts.*

Here a *causal or normative commitment* refers to the assertion of a causal relation or normative proposition as a substrate-layer fact (e.g., that event

A caused event B, or that an action constituted a violation), rather than the representation of such propositions as interpretive assertions or reports.

Proof. **(Only if.)** Suppose $\mathcal{S}$ asserts a causal or normative commitment p as a substrate-layer fact. By Lemma 4.1, p is framework-variant, and therefore there exist admissible frameworks $\mathcal{F}_1, \mathcal{F}_2 \in \mathbb{F}$ such that $\mathcal{F}_1 \vdash p$ and $\mathcal{F}_2 \vdash \neg p$. Since $\mathcal{S} \vdash p$ and $\mathcal{F}_2 \vdash \neg p$, $\mathcal{S} \cup \mathcal{F}_2 \vdash (p \wedge \neg p)$, and hence $\mathcal{S} \cup \mathcal{F}_2 \vdash \bot$. To restore consistency, either:

(i) $\mathcal{F}_2$ must be excluded from $\mathbb{F}$, violating interpretive non-commitment; or

(ii) $\mathcal{S}$ must be revised to remove p, violating extension stability.

In either case, $\mathcal{S}$ fails to satisfy neutrality.

(If.) Suppose $\mathcal{S}$ refrains from asserting causal or normative commitments at the substrate level and instead represents only framework-invariant referential structure: ontological primitives governing entity identity and persistence conditions. Under this restriction, no causal or normative proposition is fixed by the foundational layer. Let $\mathcal{F} \in \mathbb{F}$ be an admissible framework. Since $\mathcal{S}$ asserts only framework-invariant propositions, and $\mathcal{F}$ is internally consistent, no proposition asserted by $\mathcal{S}$ is contradicted by $\mathcal{F}$. Therefore, $\mathcal{S} \cup \mathcal{F} \nvdash \bot$. This holds even when role predicates or contextual typing are used at higher organizational layers, provided such predicates are not asserted as substrate-layer facts. As shown in Example 4.1, reified assertions attributed to interpretive frameworks satisfy this condition. Thus, for all $\mathcal{F} \in \mathbb{F}$, $\mathcal{S} \cup \mathcal{F}$ remains consistent, satisfying both interpretive non-commitment and extension stability.

Scope note. This direction assumes that identity and persistence conditions are not themselves contested across admissible frameworks. In domains where identity is interpretively contested, for example, where frameworks disagree about whether two records denote the same entity, or whether a particular category should be treated as a distinct entity, the substrate, under the neutrality definition, cannot be neutral until identity conditions are resolved. Such resolution is a precondition for neutrality, not a product of it. The theorem applies to domains where identity conditions can be fixed prior to interpretive extension; it does not claim that all domains admit neutral substrates. □

This result is not contingent on particular modeling choices; it follows directly from the functional role the substrate is required to play under the neutrality definition. Neutrality across disagreement and embedded interpretive commitments are structurally incompatible.

The next section considers the implications of this result for ontology design, clarifying what must be excluded from foundational layers and how interpretation, evaluation, and explanation may be externalized without loss of accountability.

5 Implications for Ontology Design

The result established in Section 4 does not prescribe a particular ontology. Rather, it constrains the design space of any ontology intended to function as a neutral substrate. This section clarifies what such constraints require, what they exclude, and how interpretation and evaluation may be accommodated without violating neutrality.

5.1 What Must Be Excluded from the Substrate

The primary implication is negative but precise. A neutral ontological substrate must exclude, at the foundational level, assertions whose truth is framework-variant across admissible frameworks. In particular, the substrate must not embed:

- normative commitments, such as obligations, permissions, prohibitions, violations, compliance determinations, or evaluative judgments that presuppose a particular legal, political, or analytic framework; and
- causal commitments, such as assertions that one event caused, produced, or was responsible for another.

These exclusions do not deny the importance of such concepts for accountability. Rather, they recognize that embedding them as ontological facts collapses the distinction between representation and interpretation, thereby violating neutrality.

The exclusions apply specifically to asserted conclusions, not to the entities or structures that such conclusions refer to. Actors, events, institutional artifacts, jurisdictions, and records may all be represented without committing to how they are interpreted or evaluated. The substrate may therefore be rich in entities and identity structure while remaining silent on contested interpretations.

5.2 What the Substrate Must Provide

Although the constraints are restrictive, they do not leave the ontology impoverished. On the contrary, a neutral substrate must make strong, explicit commitments in areas that are invariant across interpretive frameworks. In terms of the Method of Levels of Abstraction (LoA) (Floridi 2008), the substrate fixes a framework-invariant referential structure: predicate symbols governing identity and persistence conditions of entities that remain constant regardless of the explanatory logic applied to them.

The claim that a neutral substrate relies on such framework-invariant structure does not assert that all questions of entity individuation, persistence, or boundary demarcation are framework-independent. Many such questions (e.g., the spatial or temporal granularity of an event, the criteria for organizational continuity, or the recognition of informal institutions) are legitimately contested across admissible frameworks. Neutrality instead requires that the substrate fix only a minimal and domain-appropriate set of identity and persistence conditions sufficient to support accountability relations, while deferring finer-grained individuation disputes to interpretive layers. These substrate-layer commitments are invariant not because they resolve all ontological disagreement, but because they remain stable under extension by mutually incompatible causal, legal, or normative frameworks.

In particular, the substrate must provide:

- stable reference to entities that participate in accountability relationships;
- clear identity criteria and disjointness conditions for those entities;
- persistence conditions that allow entities to be tracked across time, jurisdictions, and datasets.

These commitments are not optional. Accountability depends on the ability to refer unambiguously to who acted, what occurred, which institutional instruments existed, and within what jurisdictional scope. Such commitments do not vary with interpretive stance in the way causal or normative conclusions do, and therefore do not threaten neutrality.

5.3 Externalizing Interpretation Without Loss

A common concern is that excluding causal and normative commitments from the substrate renders the ontology incapable of supporting explanation, evaluation, or judgment. The result shows that the opposite is true: such functions must be externalized in order to be supported robustly.

Interpretation is accommodated by organizing the system into distinct Levels of Organization (LoO) (Floridi 2011). The foundational LoO (the substrate) records the what (entities and identity), while higher-order LoOs record the how and why (interpretive claims). By maintaining this structural separation, multiple, mutually inconsistent interpretive LoOs can reside atop the same foundational LoO without introducing inconsistency at the substrate layer. Disagreement, revision, and contestation are then represented by the coexistence of multiple such records, rather than by revision of substrate-layer facts.

Similarly, causal and normative reasoning may be carried out within interpretive frameworks layered atop the substrate. These frameworks may draw on the same underlying entities and events while reaching incompatible conclusions. The substrate remains unchanged, serving as a stable point of reference rather than a site of resolution.

This architecture does not diminish the importance of causal reasoning; it relocates it to interpretive layers, where model commitments are explicit and conclusions may legitimately vary across admissible frameworks.

This structural separation is independent of any specific representational technology, but imposes a clear constraint on admissible designs. Implementations must ensure that causal and normative propositions are not asserted as substrate-layer commitments, but are instead represented as interpretive assertions attributed to particular interpretive contexts.

This condition may be realized through multiple mechanisms, including contextual modules (e.g., named graphs), reified assertion objects with provenance, or linked interpretive ontologies layered above the substrate. In each case, the substrate records entities and identity structure, while causal and normative claims are externalized as interpretive assertions associated with specific frameworks.

5.4 Permitted Structural Relations and Excluded Commitments

The exclusion of causal and normative primitives does not entail the exclusion of all structured relations. A neutral substrate permits relations that establish structural, temporal, or institutional linkage without asserting causal efficacy or evaluative force. Relations such as containment, delegation, temporal occurrence, participation, and applicability are admissible insofar as they do not assert that one entity caused, justified, or obligated another. By contrast, relations that embed causal attribution (e.g., X caused Y) or deontic judgment (e.g., X violated Y, X was obligated to do Y) are excluded from the substrate and must be represented, if at all, via reification or within interpretive frameworks. The neutrality boundary is thus drawn not at the presence of structure, but at the embedding of framework-dependent conclusions as ontological facts.

5.5 Reification as a Boundary Mechanism

Reification plays a critical role in enforcing the boundary between substrate and interpretation. By treating claims, reports, and assertions as entities in their own right, the ontology can represent discourse about the world without asserting conclusions about the world itself.

However, reification functions as a boundary mechanism only when applied consistently. If causal or normative relations are asserted as substrate-layer facts and merely duplicated as reified claims, neutrality is not preserved. Reification supports neutrality only when the substrate refrains entirely from endorsing the reified content.

Properly implemented, reification transforms a framework-dependent predicate (e.g., $caused(A, B)$) into an entity (e.g., $Assertion72$). This shift in type allows the substrate to maintain its identity-centric LoA while providing the raw materials for higher-level LoOs to perform their evaluative functions.

This observation reinforces the central conclusion: neutrality is not achieved by representational cleverness alone, but by disciplined exclusion of interpretive commitments from the foundational layer.

5.6 Summary of Structural Constraints

The structural constraints established by the theorem can be summarized succinctly. Any ontology satisfying the neutrality requirements must:

- restrict the foundational LoA to framework-invariant referential structure: predicate symbols governing entity identity and persistence conditions;
- assert no framework-variant proposition at the substrate layer in particular none whose truth conditions depend on causal or normative interpretation, regardless of current consensus among admissible frameworks;
- preserve the LoO hierarchy so that interpretation, evaluation, and explanation are supported exclusively through externalized higher-level organizational layers that extend but do not revise the substrate;
- satisfy extension stability by remaining consistent under extension by mutually incompatible interpretive frameworks, without revision of substrate-layer commitments.

These constraints do not define an ontology, but they sharply delimit what any such ontology can be. The interaction between these constraints is illustrated in Example 4.1.

The following section concludes by situating these constraints as boundary conditions on any ontology intended to function as a neutral substrate for accountability, independent of domain or application.

6 Conclusion

This paper has argued that ontological neutrality, when understood as interpretive non-commitment and extension stability across incompatible frameworks, imposes strict and unavoidable constraints on the design of ontological substrates. Any ontology intended to function as a neutral substrate for accountability under persistent interpretive disagreement must therefore be pre-causal and pre-normative at the substrate's foundational layer.

The argument supporting this claim is structural rather than empirical. When causal or normative commitments are asserted as substrate-level facts, neutrality is violated: the substrate cannot remain compatible with all admissible frameworks without revision or contradiction. This incompatibility arises not from disagreement itself, but from embedding contested commitments at the foundational layer rather than externalizing them to interpretive extensions.

The resulting structural constraint has both negative and positive implications. This constraint restricts what may be asserted at the foundational layer, but enables accountability by preserving stable referential structure governing the identity and persistence of entities across time and changing interpretations. Interpretation and explanation are not eliminated; they are relocated to layers where disagreement can be represented explicitly rather than suppressed through premature ontological commitment.

The result established here is conditional rather than universal. Neutral substrates are achievable only in domains where identity and persistence conditions can be fixed in a manner invariant across admissible frameworks prior to interpretive extension. Where identity itself remains irreducibly contested, neutrality, under the definition used in this paper, is not achievable, as the substrate cannot be stabilized without already committing to a particular interpretive stance. This limitation is not a defect of the framework but an explicit boundary condition on its applicability.

This result clarifies both the power and the limits of ontological neutrality.

Formally, it establishes a necessary and sufficient condition for an ontology to function as a neutral substrate under the neutrality definition used in this paper: any ontology satisfying these constraints can function as a neutral substrate in the sense defined here, while any ontology violating them cannot.

Conceptually, it delineates the space in which ontological representations can remain neutral in the presence of persistent interpretive disagreement across institutional and analytic contexts.

Limitations

The extension-stability condition analyzed here represents a boundary case: it applies specifically to substrates designed to remain stable under extension by incompatible interpretive frameworks without coordinated governance or negotiated identity criteria. Practical systems may relax this requirement through such coordination or negotiated identity criteria. In such settings, the result need not apply. Additionally, the theorem presupposes that identity and persistence conditions are invariant across admissible frameworks. In domains where identity itself is contested, neutrality in the sense defined here is unattainable until identity and persistence conditions are resolved at a prior stage.

Future Work

This structural characterization provides a basis for examining how neutral-substrate architectures can be realized in concrete domains. Demonstrating how these constraints apply in domains requiring accountability and interoperability is an important direction for future work. Exploring how existing upper ontologies might be profiled or extended to satisfy these constraints offers a promising path toward practical neutral substrates.

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The author was solely responsible for the conception, analysis, and writing of this manuscript.

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Appendix A. Glossary of Terms

This appendix defines key terms as used in this paper, organized into three categories: core ontology concepts, neutrality and interpretation, and causal and normative concepts.

Core Ontology Concepts

General ontological vocabulary used in the formal framework.

Boundary Condition

A precondition that must hold for a framework or result to apply. In this paper, the central boundary condition is that entity identity and persistence conditions must be invariant across admissible frameworks within the domain. Domains where identity is itself contested do not admit neutral substrates in the sense defined in this paper. Boundary conditions delimit scope; they are not defects but explicit constraints on applicability.

Commitment

A proposition asserted by an ontology as holding at the substrate level, independently of any particular interpretive framework. Commitments are the content of ontological assertions: what the ontology treats as true, not merely what has been claimed by external agents or frameworks. The central constraint of this paper is that causal and normative propositions may not appear as substrate-layer commitments. Contrast with *Conclusion*.

Conclusion

A proposition derived by an interpretive framework from substrate-layer commitments together with the framework's own rules and assumptions. Conclusions belong to interpretive layers, not to the substrate. Contrast with *Commitment*.

Embedding

The act of asserting a claim as a foundational fact within an ontology, such that it becomes part of the substrate-layer representation. Embedded content is treated as true by the ontology itself, not merely recorded as a claim made by some agent. Contrast with *Reification*.

Identity Conditions

The criteria that determine when two references denote the same entity and what changes an entity can undergo while remaining the same entity. Identity conditions govern individuation within the ontology.

Identity-and-Persistence Regime

A structured set of identity and persistence conditions that determine how a class of entities can be stably referenced over time within an ontology. An identity-and-persistence regime specifies what counts as the same entity, what changes are compatible with continued identity, and how references to that entity remain stable across datasets, jurisdictions, or observational contexts. In a neutral substrate, such regimes provide the identity and persistence structure upon which interpretive frameworks operate.

Level of Abstraction (LoA)

A perspective defined by a specific set of predicate symbols used to represent a domain. The LoA determines what can be represented or queried; selecting a restricted LoA excludes certain predicates from the ontology's vocabulary.

Level of Organization (LoO)

The architectural layer or scale at which entities and relationships are modeled within a system. Levels of Organization are structural rather than scalar: they differ by direction of dependence (e.g., foundational versus interpretive), not by degree, granularity, or complexity. A system may have multiple LoOs arranged hierarchically, with foundational layers supporting higher-order interpretive layers.

Ontological Assertion

The act of embedding a commitment in an ontology as a foundational fact. See *Commitment*, *Embedding*.

Ontological Primitive

A term or predicate that an ontology takes as undefined, using it to define or constrain other terms rather than deriving it from prior definitions. In a neutral substrate, the admissible primitives are restricted to those governing entity identity and persistence conditions, while causal and normative predicates are excluded from the foundational layer.

Ontology

A formal representation of entities within a domain and the relationships that hold between them, expressed in a logic or schema suitable for inference.

Ontology, Applied

The use of ontological methods to structure and analyze real-world domains for practical purposes. Applied ontology is the field; an applied ontology is an ontology deployed in a specific domain context.

Ontology, Domain

An ontology that represents the entities, relations, and constraints specific to a particular subject area, such as legal institutions, biomedical processes, or financial instruments. Domain ontologies are typically built by extending a foundational or upper ontology with domain-specific vocabulary. Contrast with *Ontology, Foundational*.

Ontology, Epistemic

An ontology that represents knowledge about the world rather than the world directly. Epistemic ontologies encode claims, beliefs, or assertions as reified entities, allowing representation of diverse perspectives without embedding contested conclusions as substrate-layer commitments. Contrast with *Ontology, Metaphysical*.

Ontology, Foundational

An ontology that provides domain-independent upper-level categories intended to be extended for specific applications. Foundational ontologies define what kinds of things exist (e.g., continuants, occurrents, roles, processes) without committing to domain-specific content. The Basic Formal Ontology (BFO 2.0) is a realist foundational ontology grounded in scientific practice, distinguishing continuants (entities that persist through time) from occurrents (events and processes that unfold over time). The Descriptive Ontology for Linguistic and Cognitive Engineering (DOLCE) takes a cognitive and linguistic orientation, emphasizing conceptualization over realism and supporting situation-based modeling of descriptions and contexts. Also called *Ontology, Upper*. Contrast with *Ontology, Domain*.

Ontology, Metaphysical

An ontology that represents the world directly, asserting what exists and what relations hold as ontological facts. Metaphysical ontologies embed commitments about the structure of reality at the substrate layer. Contrast with *Ontology, Epistemic*.

Ontology, Upper

See *Ontology, Foundational*.

Persistence Conditions

The criteria that determine how an entity continues to exist through change over time. Persistence conditions govern entity tracking across datasets, jurisdictions, and temporal contexts, and are established at the substrate layer prior to interpretive extension.

Predicate Symbol

A symbol in the ontology's formal vocabulary that denotes a property or relation. The Level of Abstraction of a substrate is identified with the set of predicate symbols available at the foundational layer. Neutrality requires that this set exclude predicate symbols whose extension depends on causal or normative interpretation.

Reification

The representation of a claim, assertion, or relation as an entity rather than as a direct ontological fact. Reification allows the ontology to represent discourse about the world without asserting conclusions about the world itself, for example, by representing that an agent asserted a causal or normative claim without embedding that claim as a substrate-layer fact. See also *Ontology, Epistemic*. Contrast with *Embedding*.

Representation

The encoding of information within an ontology. Representation is broader than embedding: an ontology may represent a claim (via reification) without embedding it as a foundational fact. See also *Embedding*, *Reification*.

Substrate

The foundational layer of an ontology that functions as a shared representational base across interpretive frameworks. Formally, the *substrate ontology* is the ontology whose foundational layer provides entities together with identity and persistence conditions. For brevity, this paper refers to that ontology simply as the *substrate*. A neutral substrate excludes causal and normative primitives, serving as a stable base for divergent interpretive layers. Also called *substrate layer* or *foundational layer*.

Vocabulary

The set of symbols available in an ontology for representing entities, properties, and relations. In a formal ontology, the vocabulary consists primarily of predicate symbols together with any constants or functions used in the representation. At a given LoA, the vocabulary determines which distinctions can be expressed.

Neutrality and Interpretation

Terms specific to the neutrality requirements defined in this paper.

Accountability System

A data system designed to support explanation, evaluation, and assignment of responsibility across multiple stakeholders or interpretive perspectives. Accountability systems must accommodate persistent disagreement about causes, obligations, and interpretations.

Admissible Framework

An interpretive framework that is internally consistent and represents a legitimate stance within the domain of application, such as a recognized legal jurisdiction, an established causal modeling tradition, or a coherent normative theory. Admissible frameworks need not be mutually compatible.

Commitment

In the neutrality context, the key distinction is between substrate-layer ontological commitments, which must remain invariant, and framework-level conclusions, which may vary. See *Commitment* in Core Ontology Concepts.

Compatibility

The property of a substrate ontology and an interpretive framework such that their combination does not entail contradiction. Formally: $\mathcal{S} \cup \mathcal{F} \nvdash \bot$.

Consistency ($\nvdash \bot$)

The property of a set of statements such that no contradiction can be derived from its assertions. The notation $\mathcal{S} \nvdash \bot$ indicates that a substrate ontology $\mathcal{S}$ is consistent; $\mathcal{S} \vdash \bot$ indicates that $\mathcal{S}$ is inconsistent (i.e., both p and $\neg p$ can be derived for some proposition p).

Extension Stability

The requirement that a substrate remain logically consistent when extended by admissible, including mutually incompatible, interpretive frameworks. Violation occurs when adding a framework forces revision of substrate-layer assertions. Together with *Interpretive Non-Commitment*, defines *Neutrality*.

Externalization

The architectural practice of representing causal or normative claims within interpretive frameworks layered above the substrate, rather than embedding them as substrate-layer commitments. Externalization preserves neutrality by ensuring that contested conclusions remain within the interpretive layers where framework assumptions are explicit and disagreement is representable. Reification is one mechanism for externalization. Named graphs, contextual modules, and linked interpretive ontologies provide alternative implementation strategies. See *Reification*.

Framework-Invariant Referential Structure

Elements of the substrate's vocabulary whose interpretation remains constant across admissible interpretive frameworks. In this paper, framework-invariant referential structure consists of predicate symbols governing entity identity and persistence conditions. Such predicates may appear as substrate-layer commitments because their interpretation does not depend on a particular interpretive framework. Contrast with *Framework-Variant Proposition*.

Framework-Variant Proposition

A proposition whose truth value may differ across admissible interpretive frameworks. Framework-variant propositions may not appear as substrate-layer commitments under the neutrality definition. Formally defined in Definition 3.2. Contrast with *Framework-Invariant Referential Structure*.

Interpretive Framework

A coherent set of principles, rules, or assumptions used to evaluate, explain, or assign meaning to entities and events. Examples include legal jurisdictions, causal models, ethical theories, and analytic methodologies. Interpretive frameworks may disagree with one another while each remaining internally consistent.

Interpretive Non-Commitment

The requirement that a substrate refrain from asserting causal or normative conclusions as substrate-layer commitments, since their truth may vary across admissible interpretive frameworks. An ontology satisfies interpretive non-commitment if it does not commit the substrate to conclusions whose truth depends on a particular interpretive framework. Together with *Extension Stability*, defines *Neutrality*.

Neutral Substrate

A foundational ontology that satisfies neutrality, enabling stable use across multiple, potentially incompatible interpretive frameworks. A neutral substrate excludes causal and normative commitments from the foundational layer, supporting pluralism and accountability.

Neutrality

The property of an ontology that avoids embedding causal, deontic, or normative commitments at the foundational layer, enabling stable use across persistent interpretive, legal, and analytic disagreement. Neutrality is defined by two requirements: interpretive non-commitment and extension stability.

Persistent Interpretive Disagreement

The condition in which multiple admissible interpretive frameworks reach incompatible conclusions about the same entities or events, and this disagreement is structural rather than resolvable by additional data or coordination. Persistent interpretive disagreement is the motivating condition for neutral substrate design. Contrast with *Semantic Heterogeneity*.

Pluralism

The coexistence of multiple, potentially incompatible interpretive frameworks within a single system. A neutral substrate supports pluralism by providing a shared foundation that does not commit the substrate to any single framework.

Semantic Heterogeneity

The condition in which multiple ontologies exist for the same domain and disagreement is treated as a translation problem addressed through alignment, mapping, or mediation. Semantic heterogeneity assumes disagreement is resolvable; persistent interpretive disagreement does not. Contrast with *Persistent Interpretive Disagreement*.

Shared Referent

An entity, event, or institutional artifact to which multiple interpretive frameworks refer using the substrate's identity and existence conditions. Shared referents are what the substrate's foundational commitments are about; they provide the common ground on which incompatible interpretive conclusions can be recognized as genuinely in conflict rather than merely equivocal.

Substrate Revision

The modification of substrate-layer commitments in response to the introduction of a new interpretive framework. Substrate revision violates extension stability and is treated in this paper as an impermissible mechanism for resolving interpretive conflict under the neutrality definition.

Causal and Normative Concepts

Terms relating to the classes of commitment excluded from neutral substrates.

Causal Attribution

The assignment of a causal relation between events, entities, or actions within an interpretive framework. Causal attribution is typically framework-dependent: since different causal models may attribute causal influence differently. Contrast with *Causal Commitment*.

Causal Commitment

An ontological commitment that asserts a causal relation (e.g., A caused B) as holding in the ontology's representation. Causal commitments depend on the causal model or explanatory framework under which the relation is interpreted and may therefore vary across admissible causal frameworks. In a neutral substrate, such commitments cannot be asserted at the foundational layer.

Deontic Commitment

An ontological commitment to obligations, permissions, prohibitions, or duties as part of the ontology's representation. Deontic commitments are a subset of normative commitments and encode what ought to be done rather than what exists or occurs. They may vary across admissible normative frameworks. In a neutral substrate, such commitments cannot be asserted at the foundational layer.

Deontic Predicate

A predicate symbol whose extension is defined by obligations, permissions, or prohibitions. Deontic predicates are normative in character and are excluded from the foundational Level of Abstraction of a neutral substrate.

Framework-Dependent

A property of claims or predicates whose truth value varies across admissible interpretive frameworks. Causal and normative claims are typically framework-dependent. In the domains considered in this paper, identity and persistence conditions are assumed to remain invariant across admissible frameworks; if they are contested, neutral substrates are not attainable under the neutrality definition.

Normative Attribution

The assignment of obligations, violations, permissions, or responsibilities to an entity or action by an interpretive framework. Normative attribution is framework-dependent and therefore appears in interpretive layers rather than as substrate-layer commitments. Contrast with *Normative Commitment*.

Normative Commitment

An ontological commitment that encodes evaluative, legal, moral, or policy-based judgments as part of the foundational representation. Normative commitments may reflect ethical, legal, institutional, or procedural frameworks and therefore vary across admissible interpretive perspectives.

Pre-Causal

The property of an ontology that excludes causal primitives from its foundational layer. Pre-causal does not mean acausal or causation-denying; it means *prior to causal attribution*. A pre-causal substrate represents entities and events without asserting causal relations between them, allowing causal reasoning to occur in interpretive layers where model assumptions are explicit and disagreement is representable. Causation is externalized, not eliminated.

Pre-Normative

The property of an ontology that excludes normative primitives from its foundational layer. A pre-normative ontology represents entities and actions without asserting obligations, permissions, or evaluations, allowing normative reasoning to occur in interpretive layers.

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